

Differential Pairs

Conductor Width (W)

0.13 mm

Conductor Spacing (S)

0.2 mm

Conductor Height (H)

0.0994 mm

Applied Voltage

1 Volts

Coupled Length

2.54 mm

Signal Risetime

1 ns

Target Zdiff

100 Ohms

+/- Tolerance = 10%

Target Zdiff Plus

112.500 Ohms

Target Zdiff Minus

87.500 Ohms

Formula Restrictions:

$0.1 < W/H < 3.0$

$0.1 < S/H < 3.0$

Differential Protocol

Ethernet

Send to Via Calculator

Zdifferential

100.221 Ohms

Zo

53.857 Ohms

Zodd

50.110 Ohms

Zeven

57.883 Ohms

? Help

Kb (Term)

0.0360

dB

-28.866

NEXT Voltage

0.0011 V

Kb' (Unterm)

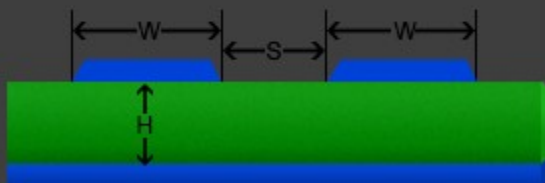
0.0720

dB

-22.857

NEXT Voltage

0.0021 V



Options

Base Copper Weight

- ☐ 9um
- ☐ 18um
- ☒ 35um
- ☐ 53um
- ☐ 70um
- ☐ 88um
- ☐ 106um
- ☐ 142um
- ☐ 178um

Plating Thickness

- ☒ Bare PCB
- ☐ 18um
- ☐ 35um
- ☐ 53um
- ☐ 70um
- ☐ 88um
- ☐ 106um

Differential Layer

- ☒ Edge CpId Ext
- ☐ Edge CpId Int Sym
- ☐ Edge CpId Int Asym
- ☐ Edge CpId Embed
- ☐ Broad CpId Shld
- ☐ Broad CpId NShld

Information

Total Copper Thickness
35 um

W/H = 1.308

S/H = 2.012

Lsat = 86.01 mm

Units

- ☐ Imperial
- ☒ Metric

Substrate Options

FR-4 STD

Er

4.1

Tg (°C)

130

Temp Rise (°C)

20

Temp in (°F) = 36.0

Ambient Temp (°C)

22

Temp in (°F) = 71.6

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Solve!



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